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Thermostable viral reverse transcriptase

Abstract

The present invention provides novel engineered reverse transcriptase enzymes that afford beneficial improvements in thermal stability, processivity, cDNA yields and elimination of secondary enzymatic activity. The present invention also provides methods for amplifying template nucleic acids using such reverse transcriptase enzymes. This invention addresses deficiencies in the current state of the art reverse transcriptase enzymes in RNA detection and analysis including deficiencies in detection sensitivity, specificity, side enzyme activities, enzyme stability and synthesis capacity, especially when using template nucleic acids ranging in length, secondary structure and nucleotide content.

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Background/Summary

(1) This application includes a Substitute Computer Readable Form (CFR) copy of a sequence listing, which is hereby incorporated by reference. The Substitute Computer Readable Form (CFR) copy of the sequence listing was submitted on Jul. 20, 2022 via EFS-web as an ASCII text file named "QIA-52303-Substitute-Sequence-Listing.txt". The ASCII text file was created on Jul. 20, 2022, and is 72 KB in size.

FIELD OF THE INVENTION

(2) The present invention provides novel engineered reverse transcriptase enzymes for improved detection and analysis of RNA. The present invention also provides methods for amplifying template nucleic acids using such reverse transcriptase enzymes and methods of producing said enzymes. This invention addresses deficiencies of current state of the art reverse transcriptase enzymes in RNA detection and analysis including deficiencies in detection sensitivity, specificity, side enzyme activities, enzyme stability and synthesis capacity, especially when using template nucleic acids ranging in length, secondary structure and nucleotide content.

TECHNOLOGICAL BACKGROUND

(3) Detection and analysis of RNA can provide insight into important biological phenomena like gene expression, gene silencing, the presence and absence of important pathogens, particularly RNA viruses, and other medically important states. The ability to monitor presence or absence or changes in the amounts of messenger RNA, noncoding RNA and viral RNA allows diagnosis of important disease states like cancer or viral infections and generally facilitates research into biological processes. For these reasons, improvements in the state of the art in these analyses are highly sought.

(4) The most common methods of RNA analysis depend on reverse transcriptase enzymes to catalyze the synthesis of complementary DNA (cDNA) using an RNA template (Spiegelman 1971, Telesnitsky 1997), which is a critical step in several related analytic and preparative methods. For example, reverse transcription PCR (RT-PCR) (Lee 1989) and its variants quantitative RT-PCR (RT qPCR), realtime quantitative RT-PCR (RT RT-PCR) (Owarzek 1992, Paria 1993, Gibson 1996) and digital RT-PCR (RT dPCR) (Warren 2006, Sanders 2018) are fundamentally two-step processes in which cDNA is synthesized by reverse transcription and amplified by PCR. Typically, these two functions are provided by separate enzymes, a reverse transcriptase (RT), often a Moloney murine leukemia virus (MMLV) RT derivative and a thermostable DNA polymerase (Pol), typically Taq Pol. The performance of these methods depends on both activities, but improvements are most commonly sought by focusing on the RT enzyme since the PCR step is generally better established and more robust.

(5) In addition to RT-PCR, areas of RNA-centered genomics focused on transcripts, noncoding RNA, and viral genomics and metagenomics rely on variants of analytic approaches referred to collectively as RNA-Seq (reviewed in Hrdlickova 2017, Boone et al 2018). Different RNA-Seq strategies are used to address different biological questions associated with phenomena like variations in gene expression and the relative abundance of transcripts, diversity of viral genomic sequences and their evolution, intron processing and splicing and other related phenomena. In all cases, the enzyme component is critical to the generation of libraries used for sequence analysis and has a substantial impact on the overall quality and reliability of the analysis. Besides RT-PCR and RNA-Seq, cDNA synthesis has a more traditional use as a preparative method related to cloning cDNA for more focused analysis on transcripts and translated proteins (Spiegelman 1971, Burrell 1996). Optimal performance in each of these analyses places unique demands on the enzymes and the choice of RT has a strong impact on the effectiveness and reliability of the method (Okello 2010, Bustin 2015).

(6) Engineering has refined the performance of RT in specific applications. Many RTs have inherent RNase H activities that digest the RNA component of RNA/DNA hybrids (Gerard 1975). This is a positive attribute for certain applications, but interferes with other common uses (Garces 1991). There is a secondary effect of disabling RNase H activity: besides eliminating its canonical function as a nuclease, eliminating RNase H activity by mutagenesis also improves thermostability. This allows synthesis at higher temperatures, thereby facilitating analysis of highly structured RNA targets. Additional research on MMLV RT indicates that certain domains, particularly the minor groove binding track (MGBT) and connection domain (CD), play roles in interacting with template (Beard 1994). Modifying either RNase H activity, the MGBT or CD or some combination can increase template switching (Garces 1991, Svarovskaia 2000) and affinity for template and, by extension, length of product. These improvements are beneficial to varying degrees in RT-PCR, preparation of RNA-Seq libraries and cDNA cloning.

(7) Numerous native and engineered RTs have been examined to address the specialized needs of these different applications. Most common are derivatives of Moloney murine leukemia virus (MMLV) RT (Gerard 1975) or, less commonly, avian myeloblastosis (AMV) RT (Kacian 1971). Other RT enzymes have been derived from alternative retroviral enzymes (e.g., U.S. Pat. No. 7,560,117), phages (Moser 2012) and bacteria (Grabco 1996) but none fully addresses the needs for highly sensitive, highly specific, robust cDNA synthesis of targets ranging in length, secondary structure and nucleotide content. These limitations in the state of the art thus create a continuing need for improvements.

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SUMMARY OF THE INVENTION

(9) This invention provides novel engineered reverse transcriptase enzymes that afford beneficial improvements in thermal stability, length of cDNAs synthesized using an RNA template, and

elimination of secondary enzymatic activity. These improvements were incorporated into the parental equine infectious anemia virus reverse transcriptase (EIAV RT) molecule by directed mutagenesis of amino acid residues residing in thumb, connection, and RNase H domains. These mutations collectively eliminate RNase H activity and increase RT activity at elevated temperature and were identified from both rational design and random mutagenesis/screening. Combining these mutations allows significant improvement in performance including the capability to synthesize cDNAs of greater than 12 kb in length at the temperatures up to 65° C.

(10) In a first aspect, the present invention relates to a reverse transcriptase (RT) comprising two subunits, wherein the two subunits are each encoded by a variant of the polynucleotide sequence of SEQ ID NO:1, wherein the amino acid sequence encoded by the variant is at least 90% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof, and wherein the variant comprises one or more mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more amino acid exchanges relative to the amino acid sequence SEQ ID NO:2 in the minor groove binding track (MGBT) of the RT's thumb domain and/or the RT's connection domain (CD).

(11) The one or more amino acid exchanges according to this aspect of the invention can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, and 271 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, and S271Q.

(12) Additionally or alternatively, the one or more amino acid exchanges can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G or I356D or I356N, N360A, G362T or G362N, and W363K or W363N or W363I, preferably I356G, N360A, G362T and W363K.

(13) According to one embodiment, the variant of SEQ ID NO:1 further comprises one or more mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more amino acid exchanges relative to SEQ ID NO:2 in the RT's RNase H domain. The one or more amino acid exchanges according to this aspect of the invention can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 443, 470, 476, 491, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are D443G, V470F, E476Q, Q491R, R526H and K553R. In a preferred embodiment, the amino acid exchanges are D443G and/or E476Q. In a highly preferred embodiment, the amino acid exchange is D443G.

(14) According to another embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q and D443G.

(15) According to a further embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G. Alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265,

267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363N and D443G. Yet alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356D, N360A, G362T, W363K and D443G. Yet alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356N, N360A, G362N, W363I and D443G.

(16) According to a further embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, R526H and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, R526H and D443G. Alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, K553R and D443G. Yet alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, R526H, K553R and D443G.

(17) According to a further embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K, Q491R, R526H and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K, Q491R, R526H and D443G.

(18) According to a preferred embodiment, a first subunit of the two subunits comprises the complete amino acid sequence encoded by the variant of SEQ ID NO:1 and a second subunit of the two subunits comprises a proteolytic fragment of the complete amino acid sequence encoded by the variant of SEQ ID NO:1.

(19) In another aspect, the invention refers to a method for amplifying template nucleic acids comprising contacting the template nucleic acids with an RT of the invention. In one embodiment, the method is reverse transcription (RT) PCR.

(20) In a further aspect, the present invention provides a kit comprising the RT of the invention and a buffer.

(21) Another aspect of the invention relates to a polynucleotide encoding an RT of the invention. A further aspect of the invention relates to a vector comprising the polynucleotide according to the invention. In another aspect, the invention relates to transformed host cells comprising said vector.

(22) In another aspect, the invention provides an RT obtainable by expression of the polynucleotide or the vector according to the invention in a host cell. In a preferred embodiment, the host cell is *E. coli*.

(23) In yet another embodiment, the invention relates to a method of producing an RT of the invention comprising isolating the RT from the transformed host cell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will be more readily appreciated by reference to the following detailed description when being considered in connection with the accompanying drawings in which:

(2) FIG. 1 shows the results of a thermal stability assay using different RTs that have been subjected to a preincubation step at the indicated temperatures prior to the RT reaction at 42° C.

(3) FIG. 2 depicts the results of a thermal stability assay measuring cDNA synthesis at elevated temperature using different RTs. Panel (a) shows the results of cDNA synthesis reactions at 50° C. Panel (b) shows the results of 2-step RT-PCR reactions where cDNA was generated at 50° C. (upper panel) and 60° C. (lower panel).

(4) FIG. 3 shows the effect of temperature on cDNA yields in a one step RT-qPCR assay using a state of the art MMLV RT with reduced RNase H activity (EnzScript) and RNase H.sup.neg EIAV RT variants according to the invention, wherein EIAV RT V3 additionally comprises point mutations within MGBT.

(5) FIG. 4 shows the effect of temperature on the yield of cDNAs of different lengths in a 2-step RT-qPCR assay using a state of the art MMLV RT (SuperScript IV) and EIAV RT variants according to the invention.

(6) FIG. 5 shows a time course of 9 kb cDNA synthesis at 55° C. by EIAV RT V3 and the state of the art MMLV RT SuperScript IV.

(7) FIG. 6 shows the effect of temperature on cDNA yields in a 2-step RT-qPCR assay using different EIAV RT variants according to the invention.

(8) FIG. 7 shows the effect of temperature on cDNA yields in a long-range endpoint PCR assay using different EIAV RT variants according to the invention.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

(9) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art (e.g., in cell culture, molecular genetics, nucleic acid chemistry, hybridization techniques and biochemistry).

(10) In practicing the present invention, many conventional techniques in molecular biology, microbiology, and recombinant DNA may be used. These techniques are well known and are explained in, for example, Current Protocols in Molecular Biology, Volumes I, II, and III, 1997 (F. M. Ausubel ed.); Sambrook et al., 1989, Molecular Cloning: A Laboratory Manual, Second Edition, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y.; DNA Cloning: A Practical Approach, Volumes I and II, 1985 (D. N. Glover ed.); Oligonucleotide Synthesis, 1984 (M. L. Gait ed.); Nucleic Acid Hybridization, 1985 (Hames and Higgins); Transcription and Translation, 1984 (Hames and Higgins eds.); Animal Cell Culture, 1986 (R. I. Freshney ed.); Immobilized Cells and Enzymes, 1986 (IRL Press); Perbal, 1984, A Practical Guide to Molecular Cloning; the series, Methods In Enzymology (Academic Press, Inc.); Gene Transfer Vectors for Mammalian Cells, 1987 (J. H. Miller and M. P. Calos eds., Cold Spring Harbor Laboratory); and Methods in Enzymology Vol. 154 and Vol. 155 (Wu and Grossman, and Wu, eds., respectively).

(11) As used herein, the term “comprising” is to be construed as encompassing both “including” and “consisting of”, both meanings being specifically intended, and hence individually disclosed

embodiments in accordance with the present invention.

(12) The term “nucleic acid sequence”, “nucleotide sequence” or “polynucleotide sequence” as used herein refers to an oligonucleotide, nucleotide or polynucleotide and fragments and portions thereof and to DNA or RNA of genomic or synthetic origin, which may be single or double stranded and represent the sense or antisense strand. The sequence may be a non-coding sequence, a coding sequence or a mixture of both.

(13) A “variant” of a polynucleotide sequence (e.g., RNA or DNA) comprises one or more mutations within the polynucleotide sequence relative to another polynucleotide sequence wherein one or more nucleic acid residues are inserted into, deleted from and/or substituted into the nucleic acid sequence. Said one or more mutations can cause one or more amino acid exchanges within the amino acid sequence the variant encodes for as compared to another amino acid sequence (i.e. a “non-silent mutation”). Variants also include nucleic acid sequences wherein one or more codons have been replaced by their synonyms which does not cause an amino acid exchange and is thus called a “silent mutation”.

(14) The term “identity” refers to a relationship between the sequences of two or more polypeptide molecules or two or more nucleic acid molecules, as determined by aligning and comparing the sequences. “Percent identity” means the percent of identical residues between the amino acids or nucleotides in the compared molecules and is calculated based on the size of the smallest of the molecules being compared. For these calculations, gaps in alignments (if any) are preferably addressed by a particular mathematical model or computer program (i.e., an “algorithm”). Methods that can be used to calculate the identity of the aligned nucleic acids or polypeptides include those described in Computational Molecular Biology, (Lesk, A. M., ed.), 1988, New York: Oxford University Press; Biocomputing Informatics and Genome Projects, (Smith, D. W., ed.), 1993, New York: Academic Press; Computer Analysis of Sequence Data, Part I, (Griffin, A. M., and Griffin, H. G., eds.), 1994, New Jersey: Humana Press; von Heinje, G., 1987, Sequence Analysis in Molecular Biology, New York: Academic Press; Sequence Analysis Primer, (Gribskov, M. and Devereux, J., eds.), 1991, New York: M. Stockton Press; and Carillo et al, 1988, SI AM J. Applied Math. 48: 1073.

(15) In calculating percent identity, the sequences being compared are typically aligned in a way that gives the largest match between the sequences. Computer programs that can be used to determine percent identity are discussed, e.g., by Pearson (Pearson, 2013, Curr. Protoc. Bioinform. 42:3.1.1–3.1.8). Such computer algorithms are used to align the two polypeptides or polynucleotides for which the percent sequence identity is to be determined. The sequences are aligned for optimal matching of their respective amino acid or nucleotide (the “matched span”, as determined by the algorithm).

(16) Certain alignment schemes for aligning two sequences may result in matching of only a short region of the two sequences, and this small aligned region may have very high sequence identity even though there is no significant relationship between the two full-length sequences.

Accordingly, the selected alignment method can be adjusted if so desired to result in an alignment that spans at least 50 or other number of contiguous nucleotides or amino acids.

(17) The nucleic acid sequences of the present invention can be prepared using standard techniques well known to one of skill in the art. The term “encoding” or “coding” refers to the inherent property of specific sequences of nucleotides in a nucleic acid, such as a gene in chromosome or an mRNA, to serve as templates for synthesis of other polymers and macromolecules in biological processes having a defined sequence of nucleotides (i.e., rRNA, tRNA, other RNA molecules) or amino acids and the biological properties resulting therefrom. Thus a gene encodes a protein, if transcription and translation of mRNA produced by that gene produces the protein in a cell or other biological system. Both the coding strand, the nucleotide sequence of which is identical to the mRNA sequence and is usually provided in sequence listings, and non-coding strand, used as the template for the transcription, of a gene or cDNA can be referred to as encoding the protein or other

product of that gene or cDNA. A nucleic acid that encodes a protein includes any nucleic acids that have different nucleotide sequences but encode the same amino acid sequence of the protein due to the degeneracy of the genetic code. Nucleic acids and nucleotide sequences that encode proteins may include introns.

(18) The term “polypeptide” is used interchangeably with “amino acid sequence” or “protein” and refers to polymers of amino acids of any length. These terms also include proteins that are post-translationally modified through reactions that include, but are not limited to, glycosylation, acetylation, phosphorylation or protein processing. Modifications and changes, for example fusions to other proteins, amino acid sequence substitutions, deletions or insertions, can be made in the structure of a polypeptide while the molecule maintains its biological functional activity. For example certain amino acid sequence substitutions can be made in a polypeptide or its underlying nucleic acid coding sequence and a protein can be obtained with like properties. Amino acid modifications can be prepared for example by performing site-specific mutagenesis or polymerase chain reaction mediated mutagenesis on its underlying nucleic acid sequence.

(19) The term “expressed” as used herein refers to transcription and/or translation of a heterologous nucleic acid sequence within a host cell. “Production” would involve both transcription and translation. The level of expression in a host cell may be determined on the basis of either the amount of corresponding mRNA that is present in the cell, and production would be based on the amount of the desired polypeptide. For example, mRNA transcribed from a selected sequence can be quantitated by Northern blot hybridization, ribonuclease RNA protection, in situ hybridization to cellular RNA or by PCR (see Sambrook et al. (1989), supra; Ausubel et al. (1994 updated), supra). Proteins encoded by a selected sequence can be quantitated by various methods, e.g., by ELISA, by Western blotting, by radioimmunoassays, by immunoprecipitation, by assaying for the biological activity of the protein, or by immunostaining of the protein followed by FACS analysis PCR (see Sambrook et al. (1989), supra; Ausubel et al. (1994 updated), supra).

(20) The term “PCR” refers to polymerase chain reaction, which is a standard method in molecular biology for DNA amplification.

(21) “RT-PCR” relates to reverse transcription polymerase chain reaction, a variant of PCR commonly used for the detection and quantification of RNA. RT-PCR comprises two steps, synthesis of complementary DNA (cDNA) from RNA by reverse transcription and amplification of the generated cDNA by PCR. Variants of RT-PCR include quantitative RT-PCR (RT-qPCR), real-time RT-PCR, digital RT-PCR (dRT-PCR) or digital droplet RT-PCR (ddRT-PCR).

(22) In the following, the invention will be explained in more detail with reference to the accompanying figures. It will be obvious for a person skilled in the art that these embodiments and items only depict examples of a plurality of possibilities. Hence, the embodiments shown here should not be understood to form a limitation of these features and configurations. Any possible combination and configuration of the described features can be chosen according to the scope of the invention.

(23) Polymerases/Enzymes

(24) Equine infectious anemia virus reverse transcriptases (EIAV RTs) are active in vivo as a heterodimer comprising subunits of 66 kDa and 51 kDa derived from a single open reading frame. The smaller subunit of 51 kDa is a proteolytic fragment of the bigger 66 kDa subunit. Expression of the open reading frame of SEQ ID NO:1 in *E. coli* gives rise to both the 66 kDa and the 51 kDa subunits that can form homodimers (i.e., p66/p66 and p51/p51) and heterodimers (i.e., p66/p51). Some commercial preparations of RTs are mixtures of the homodimers and heterodimers. While both types of dimers are active in DNA synthesis, the heterodimer presents with higher stability and processivity than the homodimers.

(25) A preferred embodiment of the RT according to the invention thus is an enzyme preparation being enriched for heterodimers. Enrichment can be done through any method known to the skilled person, for example chromatography including Heparin, hydrophobic interaction chromatography

(HIC), and ion exchanges. An enzyme preparation is enriched for heterodimers if the content of heterodimers is at least 25% higher as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In some embodiments of this aspect of the invention, the content of heterodimers is enriched by at least 30% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In another embodiment, the content of heterodimers is enriched by at least 50% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In a preferred embodiment of this aspect of the invention, the content of heterodimers is enriched by at least 75% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In a more preferred embodiment of this aspect of the invention, the content of heterodimers is enriched by at least 100% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In a particularly preferred embodiment of this aspect of the invention, the heterodimer is purified to apparent homogeneity.

(26) In a first aspect, the present invention provides a reverse transcriptase (RT) comprising two subunits, wherein the two subunits are each encoded by a variant of the polynucleotide sequence of SEQ ID NO:1, wherein the amino acid sequence encoded by the variant is at least 90% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof, and wherein the variant comprises one or more mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 in the minor groove binding track (MGBT) of the RT's thumb domain and/or the RT's connection domain (CD).

(27) The variant of SEQ ID NO:1 can comprise an arbitrary number of codons that have been replaced by their synonyms (see Table 1) which does not cause an amino acid exchange within the amino acid sequence relative to SEQ ID NO:2. Additionally, the variant comprises at least one mutation that causes an amino acid exchange relative to the amino acid sequence of SEQ ID NO:2. In some embodiments, the variant comprises at least two mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least three mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least four mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least five mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least six mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least eight mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least ten mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2.

(28) TABLE-US-00001 TABLE 1 Genetic code Amino # acid codon Start AUG 1 Met AUG 1 Trp UGG 1 Sec (UGA) 1 Pyl (UAG) 2 Tyr UAU 2 Phe UUU 2 Cys UGU 2 Asn AAU 2 Asp GAU 2 Gln CAA 2 Glu GAA 2 His CAU CAC 2 Lys AAA AAG 3 Ile AUU AUC AUA 4 Gly GGU GGC GGA GGG 4 Ala GCU GCC GCA GCG 4 Val GUU GUC GUA GUG 4 Thr ACU ACC ACA ACG 4 Pro CCU CCC CCA CCG 6 Leu CUU CUC CUA CUG UUA 6 Ser UCU UCC UCA UCG AGU 6 Arg CGU CGC CGA CGG AGA 3 Stop UAA UAG UGA

(29) In some embodiments according to this aspect of the invention, the variant of SEQ ID NO:1 comprises one mutation that causes an amino acid exchange relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises two mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises three mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises four mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises five mutations that cause amino acid exchanges relative to the

amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises six mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises eight mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises ten mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises twelve mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2.

(30) The one or more amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 can locate to the minor groove binding track (MGBT) of the RT's thumb domain spanning amino acid residues 255-298 of SEQ ID NO:2, the RT's connection domain (CD) spanning amino acid residues 320-414 of SEQ ID NO:2, and/or the RT's RNase H domain spanning amino acid residues 438-553 of SEQ ID NO:2. In some embodiments according to this aspect of the invention, the at least one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the MGBT of the RT's thumb domain. In other embodiments according to this aspect of the invention, the at least one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's CD. In further embodiments according to this aspect of the invention, the at least one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's RNase H domain. In preferred embodiments, the at least two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's CD. In another preferred embodiments, the at least two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's RNase H domain. In particularly preferred embodiments, the at least three amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain, the RT's RNase H domain and the RT's CD.

(31) In some embodiments according to this aspect of the invention, the variant of SEQ ID NO:1 comprises one or more mutations that cause one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2. This one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 can locate to the minor groove binding track (MGBT) of the RT's thumb domain spanning amino acid residues 255-298 of SEQ ID NO:2, the RT's connection domain (CD) spanning amino acid residues 320-414 of SEQ ID NO:2, and/or the RT's RNase H domain spanning amino acid residues 438-553 of SEQ ID NO:2. In certain embodiments, this one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the MGBT of the RT's thumb domain. In other embodiments, this one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's CD. In further embodiments, this one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's RNase H domain. In preferred embodiments, the variant comprises mutations that cause two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. These two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's CD. In another preferred embodiments, these two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's RNase H domain. In particularly preferred embodiments, the variant comprises mutations that cause three amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. These three amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain, the RT's RNase H domain and the RT's CD.

(32) In some embodiments according to this aspect of the invention, the amino acid sequence encoded by the variant of SEQ ID NO:1 is at least 90% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof. In other embodiments, the amino acid sequence encoded by the variant is at least 95% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof. In preferred embodiments according to this aspect of the invention, the amino acid sequence encoded by the variant is at least 97% identical to the amino acid sequence of SEQ ID

NO:2 or a fragment thereof. An amino acid sequence that is at least 90%, at least 95% or at least 97% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof retains the biological function of the amino acid sequence of SEQ ID NO:2 or a fragment thereof. More specifically, an amino acid sequence that is at least 90%, at least 95% or at least 97% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof retains the reverse transcriptase activity of EIAV-RT.

(33) According to one embodiment, the one or more amino acid exchanges according to this aspect of the invention can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, and 271 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, and S271Q. Preferably, the EIAV RT comprises at least two of said amino acid exchanges within its MGBT. More preferably, the EIAV RT comprises at least three of said amino acid exchanges within its MGBT. In some further embodiments, the EIAV RT comprises two of said amino acid exchanges within its MGBT. More preferably, the EIAV RT comprises three of said amino acid exchanges within its MGBT. Particularly preferred, the EIAV RT comprises all of said amino acid exchanges within its MGBT.

(34) In addition or alternatively to the at least one amino acid exchange within the RT's MGBT, the one or more amino acid exchanges can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G or I356D or I356N, N360A, G362T or G362N, and W363K or W363N or W363I, preferably I356G, N360A, G362T and W363K. Preferably, the EIAV RT comprises at least two of said amino acid exchanges within its CD. More preferably, the EIAV RT comprises at least three of said amino acid exchanges within its CD. In some further embodiments, the EIAV RT comprises two of said amino acid exchanges within its CD. More preferably, the EIAV RT comprises three of said amino acid exchanges within its CD. Particularly preferred, the EIAV RT comprises all of said amino acid exchanges within its CD.

(35) In addition or alternatively to the at least one amino acid exchange within the RT's MGBT, the one or more amino acid exchanges can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 443, 470, 476, 491, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are D443G, V470F, E476Q, Q491R, R526H and K553R. According to one embodiment, the EIAV RT comprises an amino acid exchange corresponding to position D443G and/or E476Q of SEQ ID NO:2. According to one embodiment, the EIAV RT comprises one amino acid exchange corresponding to position D443G of SEQ ID NO:2. This EIAV RT has an amino acid sequence of SEQ ID NO:4 that is encoded by the nucleic acid sequence of SEQ ID NO:3 or a synonymous variant thereof. According to another embodiment, the EIAV RT comprises two amino acid exchanges corresponding to positions D443G and E476Q of SEQ ID NO:2. Said EIAV RT has an amino acid sequence of SEQ ID NO:6 that is encoded by the nucleic acid sequence of SEQ ID NO:5 or a synonymous variant thereof. In a preferred embodiment, one of the amino acid exchanges is D443G. In other embodiments, the EIAV RT comprises at least two of the aforementioned amino acid exchanges within its RNase H domain. According to further embodiments, the EIAV RT comprises at least three of said amino acid exchanges within its RNase H domain. In some further embodiments, the EIAV RT comprises two of the aforementioned amino acid exchanges within its RNase H domain. According to further embodiments, the EIAV RT comprises three of said amino acid exchanges within its RNase H domain. Particularly preferred, the EIAV RT comprises the amino acid exchanges D443G, Q491R, and R526H.

(36) In some embodiments according to this aspect of the invention, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q and D443G. Preferably, the EIAV RT comprises at least two of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to

the RNase H domain and at least one of said amino acid exchanges locates to the MGBT. More preferably, the EIAV RT comprises at least three of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain and at least two of said amino acid exchanges locate to the MGBT. According to another preferred embodiment, the EIAV RT comprises at least four of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain and at least three of said amino acid exchanges locate to the MGBT. According to a further preferred embodiment, the EIAV RT comprises at least five of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain and at least four of said amino acid exchanges locate to the MGBT. In some further embodiments, the EIAV RT comprises two of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and one of said amino acid exchanges locates to the MGBT. Preferably, the EIAV RT comprises three of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and two of said amino acid exchanges locate to the MGBT. According to another preferred embodiment, the EIAV RT comprises four of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and three of said amino acid exchanges locate to the MGBT. According to a further preferred embodiment, the EIAV RT comprises five of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and four of said amino acid exchanges locate to the MGBT. Particularly preferred, the EIAV RT comprises all of the aforementioned amino acid exchanges. The latter EIAV RT has an amino acid sequence of SEQ ID NO:8 that is encoded by the nucleic acid sequence of SEQ ID NO:7 or a synonymous variant thereof.

(37) In further embodiments according to this aspect of the invention, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G. Preferably, the EIAV RT comprises at least three of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least one of said amino acid exchanges locates to the MGBT and at least one of said amino acid exchanges locates to the CD. According to another preferred embodiment, the EIAV RT comprises at least four of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least one of said amino acid exchanges locates to the MGBT and at least one of said amino acid exchanges locates to the CD. According to further preferred embodiments, the EIAV RT comprises at least five of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least two of said amino acid exchanges locate to the MGBT and at least two of said amino acid exchanges locate to the CD. According to other preferred embodiments, the EIAV RT comprises at least six of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least two of said amino acid exchanges locate to the MGBT and at least two of said amino acid exchanges locate to the CD. According to more preferred embodiments, the EIAV RT comprises at least seven of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least three of said amino acid exchanges locate to the MGBT and at least three of said amino acid exchanges locate to the CD. According to further more preferred embodiments, the EIAV RT comprises at least eight of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least three of said amino acid exchanges locate to the MGBT and at least three of said amino acid exchanges locate to the CD. According to other more preferred embodiments, the EIAV RT comprises at least nine of the aforementioned amino acid exchanges,

wherein at least one of said amino acid exchanges locates to the RNase H domain, at least four of said amino acid exchanges locate to the MGBT and at least four of said amino acid exchanges locate to the CD. In some further embodiments, the EIAV RT comprises three of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, one of said amino acid exchanges locates to the MGBT and one of said amino acid exchanges locates to the CD. According to another embodiment, the EIAV RT comprises four of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and one or two of said amino acid exchanges locate to the MGBT and the CD respectively. According to further embodiments, the EIAV RT comprises five of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, two of said amino acid exchanges locate to the MGBT and two of said amino acid exchanges locate to the CD. According to other embodiments, the EIAV RT comprises six of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and two or three of said amino acid exchanges locate to the MGBT and the CD respectively. According to more preferred embodiments, the EIAV RT comprises seven of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, three of said amino acid exchanges locate to the MGBT and three of said amino acid exchanges locate to the CD. According to further more preferred embodiments, the EIAV RT comprises eight of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and three or four of said amino acid exchanges locate to the MGBT and the CD respectively. According to other more preferred embodiments, the EIAV RT comprises nine of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, four of said amino acid exchanges locate to the MGBT and four of said amino acid exchanges locate to the CD. Particularly preferred, the EIAV RT comprises all of the aforementioned amino acid exchanges. The latter EIAV RT has an amino acid sequence of SEQ ID NO:10 that is encoded by the nucleic acid sequence of SEQ ID NO:9 or a synonymous variant thereof.

(38) According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363N and D443G.

(39) According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363N and D443G.

(40) According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356D, N360A, G362T, W363K and D443G.

(41) According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356N, N360A, G362N, W363I and D443G.

(42) According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 443, 491 and 526 of SEQ ID NO:2,

wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, D443G, Q491R and R526H. Said EIAV RT has an amino acid sequence of SEQ ID NO:12 that is encoded by the nucleic acid sequence of SEQ ID NO:11 or a synonymous variant thereof.

(43) According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 443, 491 and 526 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, D443G, Q491R and K553R.

(44) According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 443, 470, 491, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, D443G, V470F, Q491R, R526H and K553R.

(45) According to a further embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363, 443, 491 and 526 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K, D443G, Q491R and R526H. Said EIAV RT has an amino acid sequence of SEQ ID NO:14 that is encoded by the nucleic acid sequence of SEQ ID NO:13 or a synonymous variant thereof.

(46) The following Table 2 summarizes the amino acid sequences of preferred embodiments according to the present invention and indicates one nucleic acid sequence encoding it. The skilled person knows that the amino acid sequences listed can also be encoded by nucleic acid sequences synonymous to those indicated in Table 2. The present invention also encompasses any of said synonymous nucleic acid sequences.

(47) TABLE-US-00002 TABLE 2 Preferred embodiments and their nucleic/amino acid sequences

Nucleic acid	Amino acid	Variant sequence	sequence EIAV RT	WT	SEQ ID NO: 1	SEQ ID NO: 2
EIAV RT	V1	SEQ ID NO: 3	SEQ ID NO: 4	EIAV RT	V2	SEQ ID NO: 5
SEQ ID NO: 6	EIAV RT	V3	SEQ ID NO: 7	SEQ ID NO: 8	EIAV RT	V4
SEQ ID NO: 9	SEQ ID NO: 10	EIAV RT	V5	SEQ ID NO: 11	SEQ ID NO: 12	EIAV RT
V6	SEQ ID NO: 13	SEQ ID NO: 14	V4 variant	W363N	SEQ ID NO: 15	V4 variant
I356D	SEQ ID NO: 16	V4 variant	SEQ ID NO: 17	I356N/G362N/	W363I	V5 variant
SEQ ID NO: 18	R526R/K553R	V5 variant	SEQ ID NO: 19	V470F/K553R		

(48) The EIAV RT according to the invention comprises two subunits that are both encoded by a variant of the polynucleotide sequence of SEQ ID NO: 1. Variants according to the invention are described in more detail herein above. Preferably, the RT is a heterodimer. According to some embodiments, a first subunit of the two subunits comprises the complete amino acid sequence encoded by the variant and a second subunit of the two subunits comprises a proteolytic fragment of the complete amino acid sequence encoded by the variant, i.e. a proteolytic fragment of the first subunit. The first subunit has a molecular weight of 66 kDa and the second subunit has a molecular weight of 51 kDa. Expression of the open reading frame of SEQ ID NO:1 in a host cell such as *E. coli* gives rise to both the 66 kD and the 51 kD subunits that can form homodimers (i.e., p66/p66 and p51/p51) and heterodimers (i.e., p66/p51). Some commercial preparations of RTs are mixtures of the homo- and heterodimers. While both types of dimers are active in DNA synthesis, the heterodimer presents with higher stability and processivity than the homodimers. Thus, in a preferred embodiment, the heterodimeric form of EIAV RT is purified to homogeneity.

(49) In another aspect, the invention provides an RT obtainable by expression of the variant of the polynucleotide sequence of SEQ ID NO:1 or a vector comprising said variant in a host cell. Host cells that can be used to produce the EIAV RT according to the invention are mammalian cells, insect cells, yeast cells or bacterial cells. In a preferred embodiment, the host cells are mammalian cells such as HEK 293 or CHO cells or bacterial cells such as *E. coli*. The preferred host cells are

E. coli.

(50) Another aspect of the invention relates to a polynucleotide encoding an RT of the invention. Said polynucleotide is a variant of the polynucleotide sequence of SEQ ID NO:1 as described in more detail hereinabove. A further aspect of the invention relates to a vector comprising the polynucleotide according to the invention. Said polynucleotide is a variant of the polynucleotide sequence of SEQ ID NO:1 as described in more detail hereinabove. In another aspect, the invention relates to transformed host cells comprising said vector.

(51) In yet another embodiment, the invention relates to a method of producing an RT of the invention comprising isolating the RT from the transformed host cell.

(52) This invention thus provides novel engineered reverse transcriptase enzymes that afford beneficial improvements in thermal stability, processivity and thus length of cDNAs synthesized using an RNA template, cDNA yields and elimination of secondary enzymatic activity, i.e. RNase H activity. These improvements were incorporated into the parental (EIAV RT) molecule by directed mutagenesis of amino acid residues residing in thumb, connection, and/or RNase H domains. These mutations collectively eliminate RNase H activity and increase RT activity at elevated temperature and were identified from both rational design and random mutagenesis followed by screening. Combining these mutations allows significant improvement in performance including the capability to synthesize cDNAs of greater than 12 kb in length at the temperatures up to 65° C. and enables superior performance with even the most challenging RNA samples.

(53) Methods

(54) In another aspect, the present invention refers to methods for amplifying template nucleic acids comprising contacting the template nucleic acids with an RT according to the invention.

(55) Template nucleic acids according to the present invention may be any type of nucleic acids, such as RNA, DNA, or RNA:DNA hybrids. Template nucleic acids may either be artificially produced (e.g. by molecular or enzymatic manipulations or by synthesis) or may be a naturally occurring DNA or RNA. In some preferred embodiments, the template nucleic acids are RNA sequences, such as transcription products, RNA viruses, or rRNA.

(56) In some embodiments, the method referred to herein is RT-PCR. RT-PCR may be quantitative RT-PCR (RT-qPCR), real-time RT-PCR, digital RT-PCR (dRT-PCR) or digital droplet RT-PCR (ddRT-PCR).

(57) In some preferred embodiments, the method of the invention comprises the steps of a) generating cDNA using a RT of the invention; and b) amplifying the generated cDNA using a DNA polymerase such as a Taq DNA polymerase.

(58) In some embodiments, serum albumin is added during amplification, preferably recombinant human at a concentration of 1 mg/ml.

(59) In some embodiments, the method of the invention further comprises detecting and/or quantifying the amplified nucleic acids. Quantification/detection of amplified nucleic acids may be performed, e.g., using non-sequence-specific fluorescent dyes (e.g., SYBR® Green, EvaGreen®) that intercalate into double-stranded DNA molecules in a sequence non-specific manner, or sequence-specific DNA probes (e.g., oligonucleotides labelled with fluorescent reporters) that permit detection only after hybridization with the DNA targets, synthesis-dependent hydrolysis or after incorporation into PCR products.

(60) Kits

(61) Reagents necessary to perform the method of the invention may be comprised in kits.

(62) In some embodiments, the invention relates to kits for amplifying template nucleic acids, wherein the kit comprises an RT of the invention and a buffer. Optionally, the kit additionally comprises Taq DNA polymerase and/or serum albumin. Buffers comprised in the kit may be conventional buffers containing magnesium.

EXAMPLES

(63) The invention is illustrated in the following examples.

Example 1: Expression and Purification of EIAV RT

(64) The open reading frame of SEQ ID NO:1 or a variant thereof was expressed in *E. coli* which gives rise to a 66 kDa polypeptide and a proteolytically processed fragment thereof, i.e. a 51 kDa polypeptide. The 66 kDa polypeptide has an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:2. The 66 kDa and 51 kDa polypeptides form homodimers (i.e., p66/p66 and p51/p51) and heterodimers (i.e., p66/p51). In one preparation, referred to as the “homogeneous” preparation, the heterodimeric form (p66/p51) was separated from either of the homodimeric forms (p66/p66 and p51/p51) through multiple rounds of chromatography steps including heparin, hydrophobic interaction, and ion exchange chromatography; in a second preparation, referred to as the “heterogeneous” preparation, the homodimers and heterodimers were present in approximately equal amounts. While both homodimers and heterodimers are active in DNA synthesis, homogeneous preparations of the heterodimer provide higher stability and processivity than heterogenous mixes of homodimers and heterodimers, as well as improved performance (FIGS. 1 and 2). Unless indicated otherwise, all of RT variants in the following examples were tested as homogeneous preparations of heterodimers.

Example 2: Generation of EIAV RT Mutants

(65) RNase H.sup.neg Mutations

(66) Based on homologous mutations that eliminated RNase H activity of commonly studied RTs, i.e. MMLV and HIV-1 RT (Mizuno 2010, Gerard 2002), mutation of the EIAV RT RNase H domain was expected to enhance thermal stability. Thus, RNase H.sup.neg variants (EIAV RT V1 and V2; see Table 3 below) of EIAV RT were generated by site directed mutagenesis. When samples of up to 250 ng were tested, neither EIAV RT V1 nor EIAV RT V2 showed detectable RNase H activity (data not shown).

(67) Thumb Domain Mutations

(68) The minor groove binding track (MGBT) in the thumb domain is a highly conserved structural element among retroviral RTs (Beard 1998, Bebenek 1997). Biochemical and molecular modeling studies of HIV-1 RT have revealed that this element is critical for maintaining reading frames, fidelity and processivity by increasing template-primer binding affinity (Beard 1994, Bebenek 1995). Based on sequence alignment of the thumb domain of other lentiviral RTs, five point mutations (M263V/N265K/T267N/M269A/S271Q) within MGBT (EIAV RT V3) were introduced into EIAV RT V1 by site directed mutagenesis.

(69) Random Mutations and Screening on Connection and RNase H Domain.

(70) Key mutations that improve thermostability of RTs have been identified in connection and RNase H domains (Matamoros 2013). To further improve thermostability and processivity, random mutation libraries of connection and RNase H domains were constructed and screened. Based on crystal structures of HIV-1 RT complexed with template/primer and on modeling studies (Huang 1998), a region containing 10 amino acids (355-364) in the connection domain (CD) was chosen for randomization. A CD library was generated by bridging dsDNA with a mixture of 10 ssDNA oligos containing one random amino acid at each position. A random library of RNase H domain variants was generated by error prone PCR. A screening assay of the libraries was performed by measuring RT activity of heat treated (50° C. for 10 min) crude cell lysate at 60° C. The results identified four variants from the CD library and three variants from the RNase H domain library that showed higher RT activity than the parent EIAV RT V3 (data not shown). The best performing CD variant was EIAV RT V4 and the best performing RNase H domain variant was EIAV RT V5. The mutations of EIAV RT V4 and EIAV RT V5 were combined to generate the variant EIAV RT V6.

(71) TABLE-US-00003 TABLE 3 EIAV best mode constructs SEQ Description of the Variant ID Mutation Effect EIAV SEQ none Heterodimeric wild type RT ID NO enzyme WT 2 EIAV SEQ D443G Heterodimeric enzyme RT ID NO mutagenized to eliminate V1 4 RNase H activity EIAV SEQ D443G/E476Q Heterodimeric enzyme RT ID NO mutagenized to eliminate V2 6 RNase H

activity EIAV SEQ D443G/M263V/N265K/ Heterodimeric enzyme RT ID NO T267N/M269A/S271Q mutagenized to eliminate V3 8 RNase H activity and modify the MGBT in thumb domain to increase affinity for template EIAV SEQ D443G/M263V/N265K/ MGBT mutant with RT ID NO T267N/M269A/S271Q/ additional CD mutations V4 10 I356G/N360A/G362T/ W363K EIAV SEQ D443G/M263V/N265K/ MGBT mutant with RT ID NO T267N/M269A/S271Q/ additional RNase H V5 12 Q491R/R526H domain mutations EIAV SEQ D443G/M263V/N265K/ MGBT mutant with RT ID NO T267N/M269A/S271Q/ additional CD and V6 14 I356G/N360A/G362T/ RNase H domain W363K/Q491R/R526H mutations

Example 3: Assessment of Thermostability and Thermal Activity of RNase H.SUP.neg .Variants

(72) Thermostability and thermal activity of the RNase H variants EIAV RT V1 and V2 were evaluated by thermal inactivation assay (FIG. 1) and first strand cDNA synthesis (FIG. 2). In the thermal inactivation assay, RTs were preincubated with a substrate (Oligo (dT)20 primed Poly (A) template) at the 42-60° C. for 10 min. Following preincubation, RT reaction was initiated by adding MgCl.sub.2 and the activities were measured by monitoring the relative rates of dT incorporation at 42° C. All the RNase H.sup.neg variants (EIAV RT V1 and V2) remained fully active after preincubation at 50° C. whereas heterogeneous EIAV RT WT and homogeneous EIAV RT WT showed 50 and 30% reduced activities, respectively (FIG. 1). Notably, double H.sup.neg mutant, EIAV RT V2 retained 80% of its activity even at 55° C. preincubation.

(73) The thermostability of these variants was more stringently evaluated by measuring cDNA synthesis at elevated temperature. First strand cDNA synthesis reactions were performed at 50° C. using a mixture of RNAs ranging in length from 0.5 to 9 kb as template (poly(A)-tailed RNA ladder). Single strand cDNA was resolved by alkaline electrophoresis and visualized by staining with SYBR Gold. As shown in FIG. 2(a), all RNase H.sup.neg variants efficiently synthesized full-length cDNAs up to 9 kb, while EIAV RT WT exhibited lower cDNA yield than RNase H.sup.neg variants (Lane 1: heterogeneous EIAV RT, Lane 2: homogeneous EIAV RT WT, Lane 3: EIAV RT V1, Lane 4: EIAV RT V2). In contrast, full-length products of any size from heterogeneous EIAV RT WT are near the limit of visual detection, while homogeneous EIAV RT WT yield was slightly lower than the RNase H.sup.neg variants.

(74) In a 2-step RT-PCR assay, initial cDNA synthesis reactions were done at 50° C. (FIG. 2(b), upper panel) or 60° C. (FIG. 2(b), lower panel) for 30 min using 10 ng of human total RNA as template. A 2 kb fragment of APC gene was then amplified. All RNase H.sup.neg variants generated amplifiable cDNA at both 50° C. and 60° C. whereas homogeneous and heterogeneous EIAV RT both failed to generate cDNA at either 50° C. or 60° C. (FIG. 2(b)). Taken together, these results collectively support the interpretation that abolishing RNase H activity of RT increases its thermostability.

Example 4: Assessment of Thermostability and Thermal Activity of an RNase H.SUP.neg .EIAV RT Variant with Additional Mutations within the MGBT of the Thumb Domain (EIAV RT V3)

(75) The effects of the additional mutations on thermostability and thermal activity were evaluated using a FAM probe-based one-step RT-qPCR. First, a region of the ACTB gene was reverse transcribed from human total RNA (2 pg) using EIAV RT V1, EIAV RT V3 or EnzScript RT as indicated, and then amplified and quantified. The results show that both EIAV RT V1 and EIAV RT V3 retain most of their activity close to 70° C. while showing optimal activities at temperature around 55° C. However, reduced activity of EIAV RT V1 becomes noticeable at temperatures higher than 65° C. In contrast, substantial increases of Ct values, denoting reduced cDNA yields, were observed with a typical, widely used comparator EnzScript (a Moloney murine leukemia virus (MMLV) RT with reduced RNase H activity) as temperature increases (FIG. 3).

(76) Additionally, a 2-step RT-qPCR assay was performed using EIAV RT V1, EIAV RT V3 and SuperScript IV. Using human total RNA (50 ng) and a gene specific primer (MAP4), cDNA was reverse transcribed at the indicated temperatures and cDNAs of different lengths were quantified by qPCR targeting the MAP4 gene. The results demonstrate that EIAV RT V3 generates higher

cDNA yield, as shown by lower Ct value, than EIAV RT V1. The difference in yield between these enzymes was dramatically increased at higher temperature or for longer cDNA synthesis indicating improvement of thermostability and processivity of EIAV RT V3 compared with EIAV RT V1 (FIG. 4).

(77) The speed of synthesis and length of product for cDNAs generated by the EIAV RT V3 were compared to SuperScript IV, a widely used commercial RT that is considered state of the art. The reaction was performed using 50 ng of human total RNA as input at 55° C. and Oligo (dT)20 as a primer. The reaction was quenched by heat deactivating RTs at 85° C. for 5 min at indicated times followed by amplification of a 9 kb APC gene. Both enzymes generated amplifiable 9 kb products within 10 min. However, yields were substantially higher with EIAV RT V3 (FIG. 5).

Example 5: Assessment of Thermostability and Thermal Activity of RNase

H.SUP.neg./MGBT.SUP.mut .EIAV RT Variants with Additional Mutations within the Connection Domain and/or RNase H Domain

(78) The effects of additional mutations within the CD and/or RNase H domain on thermostability and thermal activity were tested in 2 step RT-qPCR (FIG. 6) and a long-range endpoint PCR (FIG. 7).

(79) For the 2 step RT-qPCR, cDNA synthesis using human total RNA (50 ng) and a gene specific primer (MAP4) was performed at the indicated temperatures. Then, cDNA of different lengths were quantified by amplifying MAP4 gene using primer sets designed for the specific regions of the gene. For the long range 2-step RT-PCR (12.3 kb) total rat brain RNA (50 ng) and a gene specific primer for rat dynein were used for the cDNA synthesis at different temperatures (FIG. 7; Lane 1: EIAV RT V3, Lane 2: EIAV RT V4, Lane 3: EIAV RT V5, Lane 4: EIAV RT V6).

(80) Both 2 step RT-qPCR (FIG. 6) and long-range endpoint PCR (FIG. 7) assays at different temperature confirm that all of these variants (EIAV RT V4, EIAV RT V5 and EIAV RT V6) showed increased thermostability and processivity compared with the original EIAV RT V3. One of the constructs, EIAV RT V4, was capable of synthesizing 12.3 kb cDNA at temperatures as high as 65° C. (FIG. 7), and this variant is considered best mode for most applications.

Claims

1. A reverse transcriptase comprising two subunits, wherein the two subunits are each encoded by a variant of the polynucleotide sequence of SEQ ID NO:1, wherein the amino acid sequence encoded by the variant is at least 90% identical to the amino acid sequence of SEQ ID NO: 2, wherein the variant comprises mutations in the polynucleotide sequence of SEQ ID NO:1 causing amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 in the minor groove binding track (MGBT) of the reverse transcriptase's thumb domain, and wherein the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, and 271 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, and S271Q.
2. The reverse transcriptase according to claim 1, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more further amino acid exchanges relative to SEQ ID NO: 2 in the reverse transcriptase's connection domain.
3. The reverse transcriptase according to claim 2, wherein the one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are 1356G/1356D/1356N and/or N360A and/or G362T/G362N and/or W363K/W363N/W363I.
4. The reverse transcriptase according to claim 2, wherein the one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within

the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G and/or N360A and/or G362T and/or W363K.

5. The reverse transcriptase according to claim 2, wherein the one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G, N360A, G362T and W363K.

6. The reverse transcriptase according to claim 1, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more further amino acid exchanges relative to SEQ ID NO: 2 in the reverse transcriptase's RNase H domain.

7. The reverse transcriptase according to claim 6, wherein the further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 443 and 476 of SEQ ID NO:2, wherein the amino acid exchanges are D443G and/or E476Q.

8. The reverse transcriptase according to claim 6, wherein the further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause a further amino acid exchange within the encoded amino acid sequence at the amino acid position that corresponds to position 443 of SEQ ID NO:2, wherein the amino acid exchange is D443G.

9. The reverse transcriptase according to claim 1, wherein the variant comprises a further mutation in the polynucleotide sequence of SEQ ID NO:1 causing a further amino acid exchange relative to SEQ ID NO:2, wherein the further mutation in the polynucleotide sequence of SEQ ID NO:1 causes a further amino acid exchange within the encoded amino acid sequence at the amino acid position that corresponds to position 491 of SEQ ID NO:2, wherein the amino acid exchange is Q491R.

10. The reverse transcriptase according to claim 9, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 causing one or more further amino acid exchanges relative to SEQ ID NO:2, wherein the further mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 470, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are V470F and/or R526H and/or K553R.

11. The reverse transcriptase according to claim 9, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 causing one or more further amino acid exchanges relative to SEQ ID NO:2, wherein the further mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are R526H and/or K553R.

12. The reverse transcriptase according to claim 9, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 causing one or more further amino acid exchanges relative to SEQ ID NO:2, wherein the further mutations in the polynucleotide sequence of SEQ ID NO:1 cause a further amino acid exchange within the encoded amino acid sequence at the amino acid position that corresponds to position 526 of SEQ ID NO:2, wherein the amino acid exchange is R526H.

13. The reverse transcriptase according to claim 1, wherein the mutations in the polynucleotide sequence of SEQ ID NO: 1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G.

14. The reverse transcriptase according to claim 1, wherein a first subunit of the two subunits

comprises the complete amino acid sequence encoded by the variant and wherein a second subunit of the two subunits comprises a proteolytic fragment of the complete amino acid sequence encoded by the variant.

15. A kit comprising: a) the reverse transcriptase of claim 1; and b) a buffer.

16. A polynucleotide encoding an reverse transcriptase according to claim 1.

17. A vector comprising the polynucleotide according to claim 16.

18. A transformed host cell comprising the vector according to claim 17.

19. A reverse transcriptase obtained by expression of the polynucleotide according to claim 16 in a host cell.

20. The reverse transcriptase according to claim 19, wherein the host cell is *E. coli*.

21. A reverse transcriptase obtained by expression of the vector according to claim 17 in a host cell.

22. The reverse transcriptase according to claim 21, wherein the host cell is *E. coli*.

23. A method for amplifying template nucleic acids comprising contacting the template nucleic acids with an reverse transcriptase according to claim 1.

24. The method according to claim 23, wherein the method is reverse transcription (RT) PCR.
